

Bishal Das

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Summary of Qualifications

- Experience building end-to-end Machine Learning and RAG pipelines using Python, LangChain, and FastAPI
- Working knowledge of data engineering concepts, including Apache Spark, basic Apache Kafka, and ETL workflows.
- Skilled in deploying IoT and TinyML systems on embedded hardware such as ESP32

Education

ICFAI University, Tripura <i>Bachelor of Science, Computer Science and Engineering</i> Major CGPA: 9.3/10.0	Tripura, India <i>August 2024 – May 2027</i>
Tripura Institute of Technology <i>Diploma, Mechanical Engineering</i> CGPA: 7.5/10.0	Tripura, India <i>2021 – 2024</i>
Konaban (West) Class XII School <i>Higher Secondary (Class X)</i> Percentage: 62%	Konaban, Bishalgarh, Sepahijala, Tripura

Technical Skills

- **Programming Languages:** Python, SQL, C++, Java, Kotlin, Bash
- **Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, Keras, OpenCV, MediaPipe, CNN, LLMs, LangChain, LangGraph, RAG, Prompt Engineering
- **Data Engineering:** Apache Spark, PySpark, Apache Kafka, ETL, Stream Processing, Pandas, NumPy
- **Web & Backend:** FastAPI, Streamlit, REST APIs, JWT Authentication
- **Cloud & Databases:** AWS (EC2, S3, IAM), MySQL, PostgreSQL, ChromaDB
- **Tools & Technologies:** Git, GitHub, Docker, Linux

Experience

Birla Institute of Technology (BIT), Mesra <i>External Summer Intern – TinyML and Wearable Health Monitoring</i>	Ranchi, Jharkhand <i>May 2026 – July 2026</i>
• Built "A Fusion-based Human Health Monitoring System using TinyML and Wearable Sensors with ESP-32" • Designed a multi-sensor fusion pipeline for on-device health monitoring, combining wearable sensor streams for real-time inference	
National Institute of Technology (NIT) Silchar <i>AI Research Intern</i>	Silchar, Assam <i>June 2025 – July 2025</i>

- Developed an ML-based smart irrigation system, cutting water usage by 30% at 88% prediction accuracy
- Integrated a Firebase IoT backend enabling real-time sensor sync with sub-second latency

Project Experience

- **Career-Sync-AI** *AI-Powered ATS Resume Analyzer*
 - Built a Retrieval-Augmented Generation (RAG) pipeline for resume-job matching using LangChain, ChromaDB, and Mistral AI.
 - Developed a FastAPI backend and Streamlit dashboard to generate ATS compatibility scores and provide skill-gap analysis.
- **StudyRAG-AI** *RAG-Based PDF Q&A Chat Application*
 - Built a full-stack Retrieval-Augmented Generation (RAG) application for PDF question answering with OCR-based document routing and contextual compression retrieval.
 - Implemented semantic search using LangChain, ChromaDB, and FastAPI to deliver accurate, context-aware responses from uploaded documents.
 - Technologies: Python, FastAPI, LangChain, ChromaDB, Streamlit, OCR.
- **HealthGuard** *IoT Health & Activity Monitoring System*
 - Developed an ESP32-based wearable system to stream real-time sensor data via MQTT for remote health monitoring.
 - Implemented machine learning models for human activity recognition and physiological anomaly detection.

Certifications

- AI/ML Internship Certificate – National Institute of Technology (NIT) Silchar
- Project Completion Certificate – Birla Institute of Technology, Mesra